Q proves the successor commutation for numerals

The Open Logic Project contributors

openlogic:incompleteness/representability-in-q:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-incompleteness-representability-in-q/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the internal step labels (step1-step6) are the author's own ollabels and are retained

Editorial note. The label lem:succ names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Q axioms for successor and addition. Q_1: x' + y = (x + y)'; Q_4: x + 0 = x; Q_5: x + y' = (x + y)'; identity rules.. Each displayed line of the derivation is one of these axioms or an identity-rule application; the proof is a two-chain algebraic shuffle.

Lemma 0.1. *For every constant a and every natural number n,*

$$\mathrm{Th}Q \vdash (a' + \overline{n}) = (a + \overline{n})'.$$

Proof. The proof is, as usual, by induction on n. In the base case, $n = 0$, we need to show that ThQ proves $(a' + \mathsf{0}) = (a + \mathsf{0})'$. But we have:

$$\mathrm{Th}Q \vdash (a' + \mathsf{0}) = a' \quad \text{by axiom } Q_4 \tag{1}$$

$$\mathrm{Th}Q \vdash (a + \mathsf{0}) = a \quad \text{by axiom } Q_4 \tag{2}$$

$$\mathrm{Th}Q \vdash (a + \mathsf{0})' = a' \quad \text{by 2} \tag{3}$$

$$\mathrm{Th}Q \vdash (a' + \mathsf{0}) = (a + \mathsf{0})' \quad \text{by 1 and 3}$$

In the induction step, we can assume that we have shown that Th$Q \vdash (a' + \overline{n}) = (a + \overline{n})'$. Since $\overline{n+1}$ is $\overline{n}'$, we need to show that ThQ proves $(a' + \overline{n}') = (a + \overline{n}')'$. We have:

$$\mathrm{Th}Q \vdash (a' + \overline{n}') = (a' + \overline{n})' \quad \text{by axiom } Q_5 \tag{4}$$

$$\mathrm{Th}Q \vdash (a' + \overline{n}') = (a + \overline{n}')' \quad \text{inductive hypothesis} \tag{5}$$

$$\mathrm{Th}Q \vdash (a' + \overline{n})' = (a + \overline{n}')' \quad \text{by 4 and 5.}$$

□